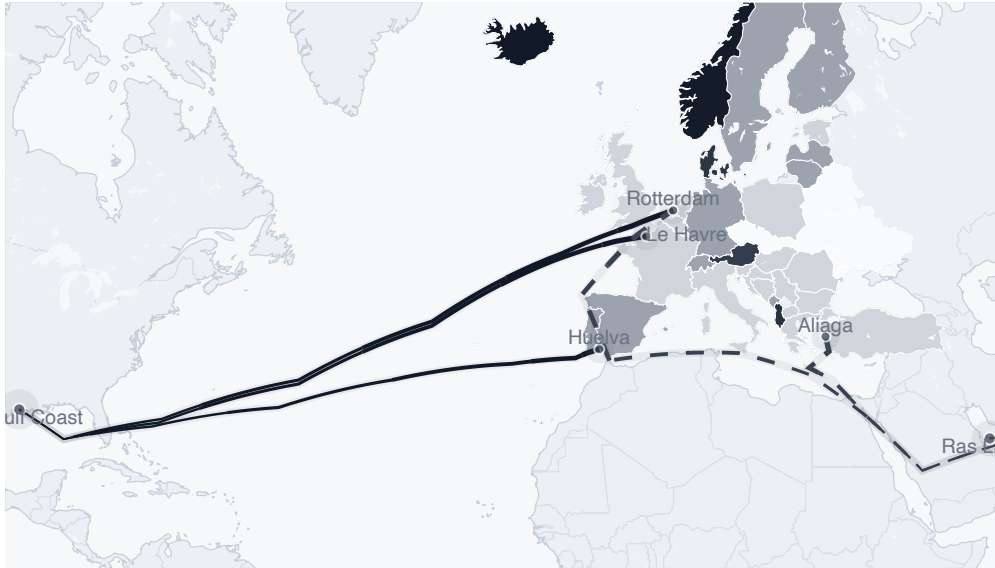


The plants that set Europe's electricity price can no longer earn a good living from doing it.

WHERE EUROPE'S GAS ARRIVES FROM



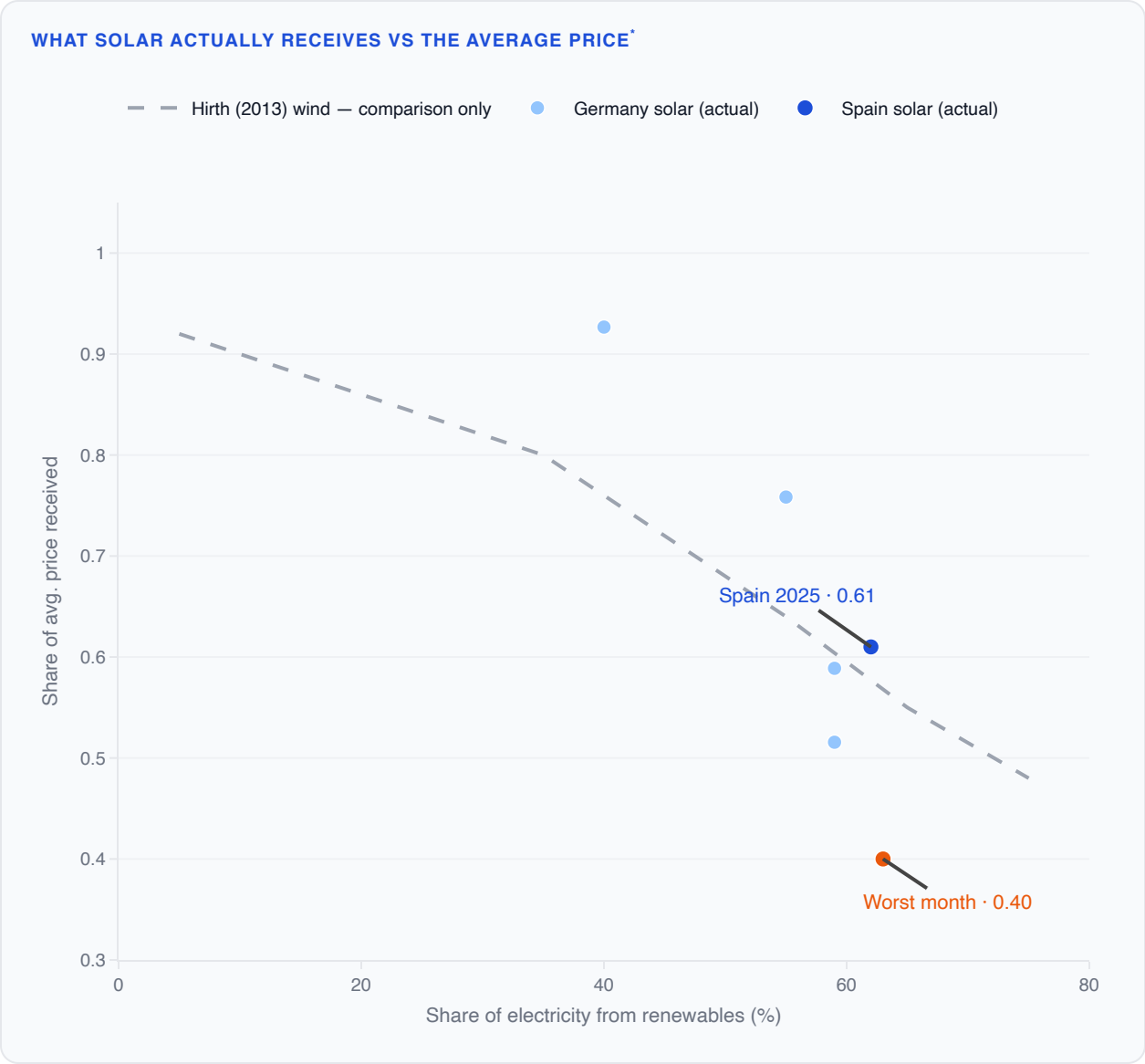
When there is a lot of sun and wind at once, solar and wind farms sell into cheap hours and undercut each other. At the same time, a wave of new gas-export terminals* is locking in cheaper gas for years ahead.

Gas shipped from the US and Qatar still matters a lot: gas-fired plants are often the last ones needed to keep the lights on, so the gas price still sets the electricity price in many hours.

In short: more renewables and cheaper gas are both pushing the same electricity price lower. The next slides show how — and what that means for people who put money into power plants.

*LNG terminals: places that ship natural gas in liquid form across the ocean.

In Spain, solar already earns only about 61 cents for every euro of the average market price.



SPANISH SOLAR · 2025

0.61

of the average price in 2025 · worst month 0.40

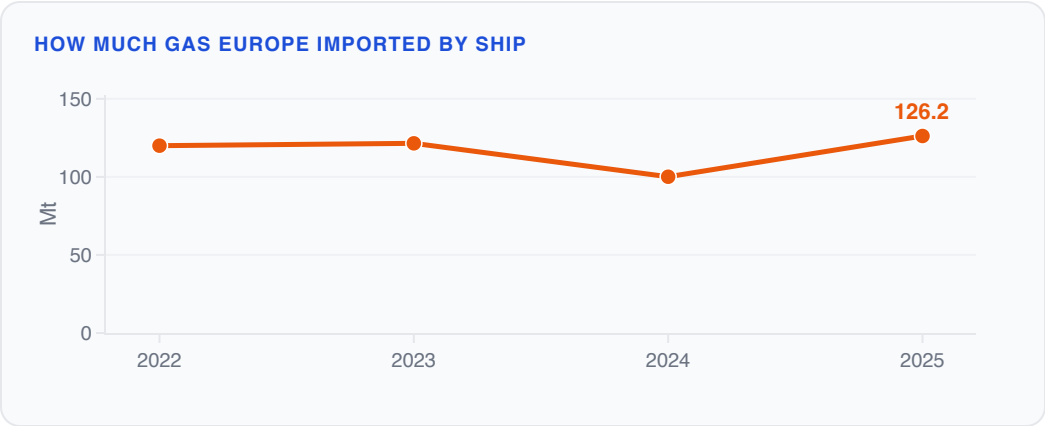
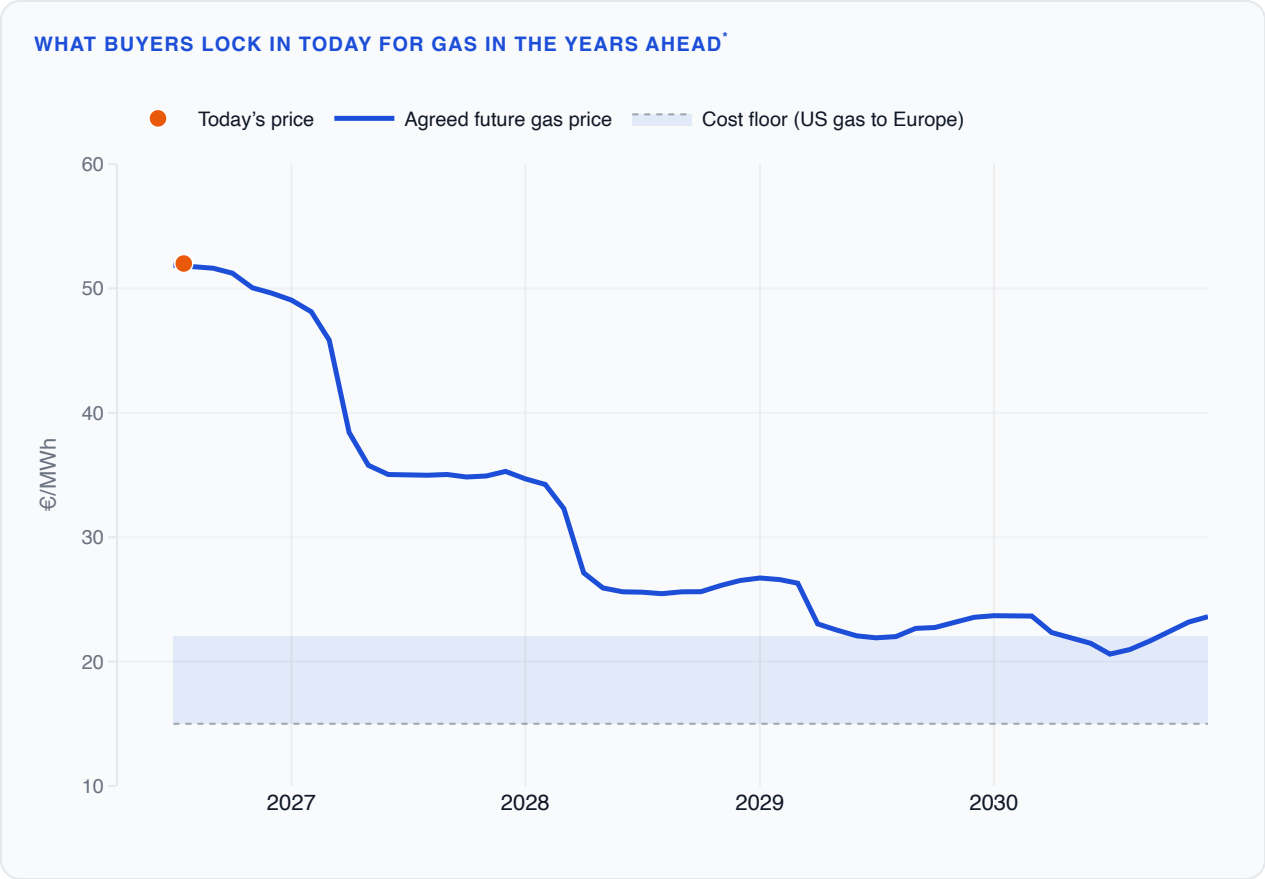
Solar produces most when the sun is out — and so does every other solar farm. In those hours the market price falls, so solar sells cheap. The dots are real market outcomes for solar in Spain and Germany; the dashed line is an older textbook curve for *wind*, shown only as a comparison.

Spain's 2025 average (0.61) and its worst month (0.40) sit well below what older business plans assumed.

In short: plans that assume solar will earn close to the full average market price are describing a market that no longer exists.

*Share of the average price received = solar's € per MWh ÷ the market's average € per MWh. Dashed line: Hirth (2013) wind curve, for comparison — not a solar forecast.

Today’s gas spikes are temporary. Markets already expect gas near €20 later this decade.



Short-term scares (a blocked shipping lane, a cold winter) can still push today’s price up. Looking further out, the market expects much lower prices — because a huge amount of new US export capacity has already been approved and is being built. Europe was the main destination for those US cargoes in 2025.

In short: by 2030, European gas is expected to settle near the cost of producing and shipping US gas (€15 to €22 per MWh[†]) — not near today’s crisis levels.

*The blue line is the main European gas benchmark (TTF): prices agreed today for delivery later. †MWh = megawatt-hour, a standard unit of energy. The shaded band is a rough cost floor for US gas delivered to Europe.

Five years on, the average electricity price is roughly half — without inventing a new crisis.



Two things happen at once on the chart: more hours where power is free or even negative, and fewer hours where gas plants set the price. Cheaper gas reinforces that second shift — but the picture here is about *who sets the price and how often*, not the running cost itself.

Average price

-34.9 €/MWh

Hours with free / negative prices

+12pp

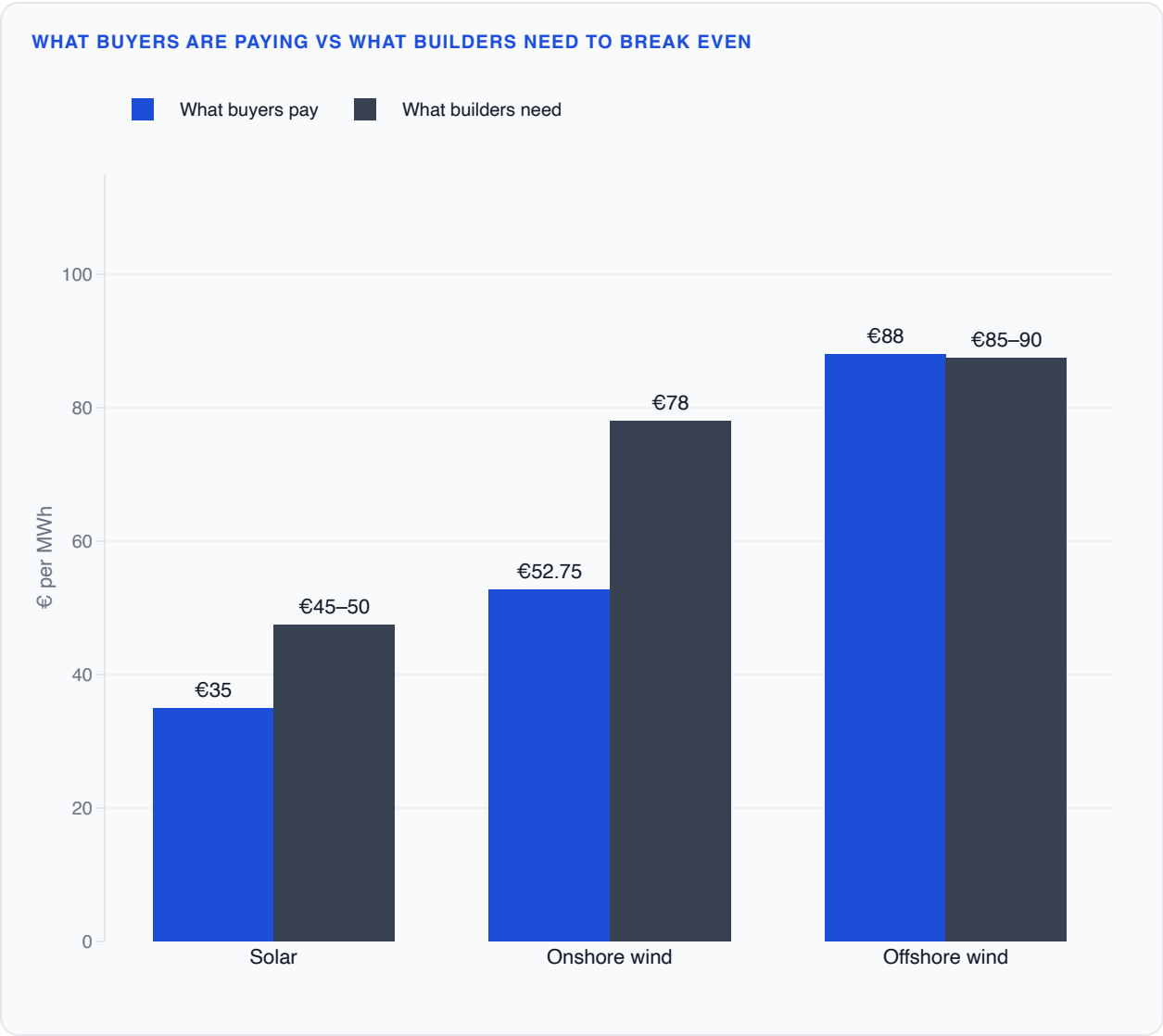
Hours when gas sets the price

-9pp

In short: more solar/wind and cheaper gas are not two stories that cancel out. They both push the electricity price down.

*In each hour, the most expensive plant still needed sets the price everyone receives.

Long-term power contracts help a project get a loan — but solar and onshore wind still clear below the cost of building.



Governments and big buyers can offer fixed-price deals* so a wind or solar farm can get a bank loan. Look at the chart: solar and onshore sit clearly below the “builders need” bar; offshore German wind only just clears it.

SOLAR
–€12.5
below what it costs to build

ONSHORE WIND
–€25.3
below what it costs to build

OFFSHORE WIND (GERMANY)
+€0.5
only just covers the cost of building

In short: solar and onshore still pay less than it costs to build; offshore is right on the edge. A fixed-price contract can rescue one project — it does not lift the open market for everyone else.

*Often called a PPA (a long-term sale of electricity) or a CfD (a government-backed price guarantee).

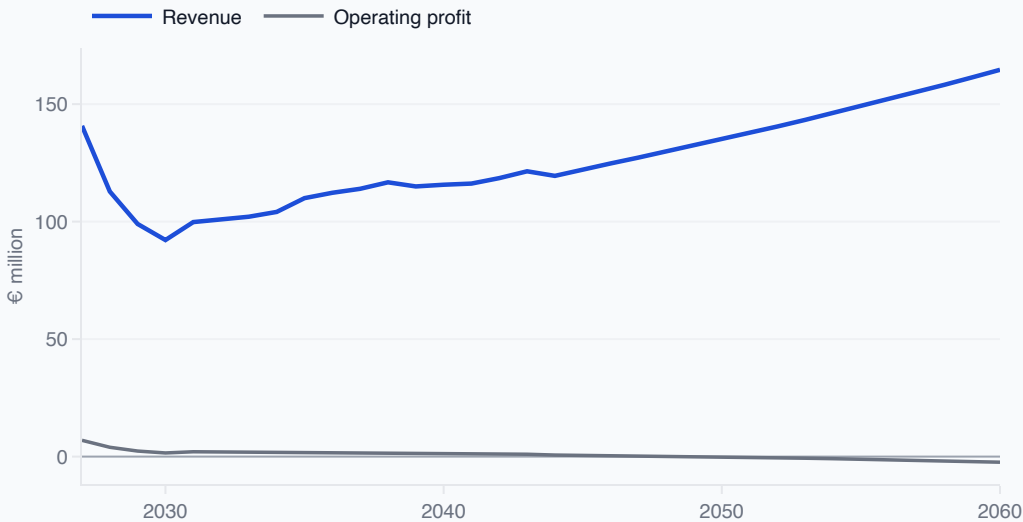
A gas plant that is already built can still be worth something. Building a new one destroys hundreds of millions in value.

ALREADY BUILT

+€25.6m

Investor return* -4.8% (vs 6% Ke)
Operating profit thin throughout; turns negative mid-century

REVENUE & OPERATING PROFIT

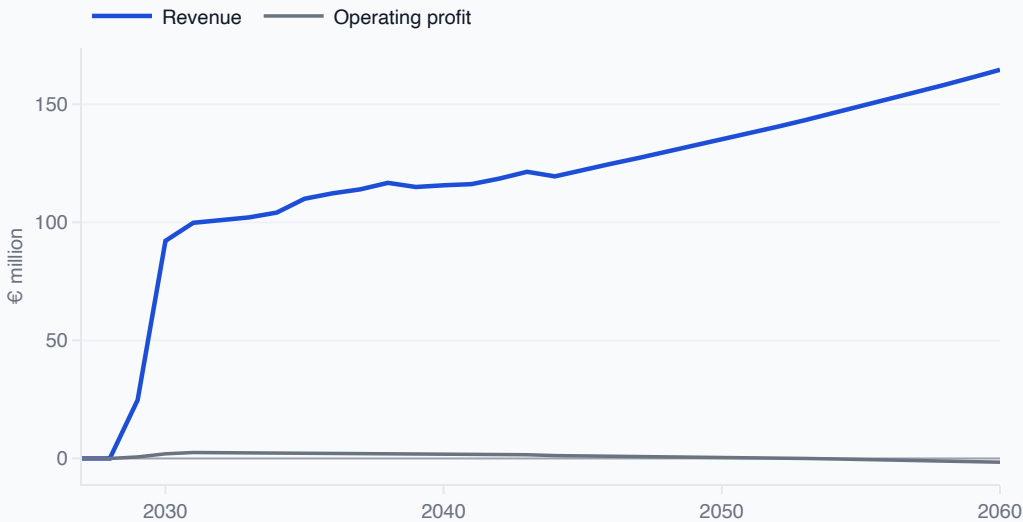


STILL TO BE BUILT

-€482.9m

Revenue path nil through construction; then tracks the operating case
Operating profit equally thin once online

REVENUE & OPERATING PROFIT



In short: once running, both plants show the same profile — revenue remains sizeable, operating profit is already thin. The existing asset can still clear a positive equity value because construction is sunk. The new build cannot recover that outlay on this price path.

Headline figures = present value of equity cash flows. *Equity IRR vs a 6% cost of equity.

Changing costs, efficiency, or how much is borrowed moves the numbers — not enough to make a new plant work.

Each row asks: if we change only this one assumption, how much cash does the owner have by 2049? The middle column is the starting point. Green = better; red = worse.

CASH TO INVESTORS BY 2049 · CHANGE ONE ASSUMPTION AT A TIME					
Running costs yearly upkeep cost	€10,000 64.4m	€11,000 51.0m	€12,000 (base) 38.9m	€13,500 24.0m	€15,000 12.6m
Plant efficiency how much energy from each unit of gas	61% -118.0m	63% 3.7m	65% (base) 38.9m	66% 82.4m	67% 137.6m
Wear and tear % performance lost each year	0.15% 90.6m	0.20% 54.8m	0.25% (base) 38.9m	0.35% 25.5m	0.50% 18.1m
If gas prices move vs today's outlook	-20% 45.8m	-10% 28.5m	+0% (base) 38.9m	+10% 51.9m	+20% 65.9m
If gas sets prices more/less often change in how often gas sets the price	-10pp 4.1m	-5pp 14.5m	+0pp (base) 38.9m	+5pp 70.6m	+10pp 105.7m
Share borrowed from banks new plant · % paid with a bank loan	40% -550.3m	50% -553.3m	60% (base) -556.3m	70% -559.4m	80% -562.4m

In short: for a plant already running, the base case is still positive (+€38.9m). For a new build, whether you borrow 40% or 80%, cash stays deeply negative. The loan terms are not the main problem — the price outlook is.

Bottom row forces a new-build case on purpose. · “Share borrowed” = how much of the project is paid with bank debt instead of investor money.